

5LSL0 Machine Learning for signal processing - assignment 3

Gheorghiu Cristian Stefan*

1 EXERCISE 3.8

1.1 3.8.a Requirement

Repeat exercise 6 using your new VAE.

1.2 3.8.a Answer

Based on the model's performance (quantified in terms of ELBO loss applied on the validation set), the network successfully reconstructs the digits, though it falls short in a few areas. For instance, it fails to perfectly match the thickness of the digit strokes, resulting in reconstructions that appear blurrier and thicker than the clean images. Additionally, the model struggles to differentiate between closely located rounded and straight contours. As Figure 1 shows, this is evident in the final digit on the right-hand side (a '4'), where the intersection of the lines blurs into an ellipsoidal shape rather than forming sharp angles.



Figure 1: Top-Bottom comparison of the input, VAE output and the ground truth label

This limitation may stem from the heavily skewed loss weighting adopted during training ($\beta = 0.005$), where $total_{loss} = recon_{loss} + \beta \times KL_{divergence}$. By drastically down-weighting the KL divergence, the model focuses almost entirely on the reconstruction quality while largely ignoring whether the latent distribution resembles the prior distribution. This demonstrates that when the model operates away from the global optimum, the true, intractable log-likelihood remains strictly greater than our tractable surrogate objective. Consequently, $q_V(z|x_i) \neq p_\theta(z|x_i)$, indicating that the Evidence Lower Bound (ELBO) is not tight and we are failing to maximize the true marginal likelihood. When employing the standard ELBO loss (a direct summation of the reconstruction error and the KL divergence), the results remain largely unchanged, at least within the first 30 epochs. At this stage, the model is evidently still converging, as the validation loss continues to steadily decrease from one epoch to the next.

1.3 3.8.b Requirement

Now, implement equation (4) using the decoder of your trained VAE. We can find the argmin through an optimization procedure. This time not on the parameters of a network, but on the values of the latent space. Using the noisy measurements of the 10 examples, try to create clean images using this algorithm. Similar to 8a, show the

*e-mail: c.gheorghiu1@student.tue.nl

noisy measurements you used as input, together with the reconstructions and the ground truth for the example images. Moreover, also show a plot of the MAP loss over the iterations. For implementation hints, take a look at MAP-template.py

1.4 3.8.b Answer

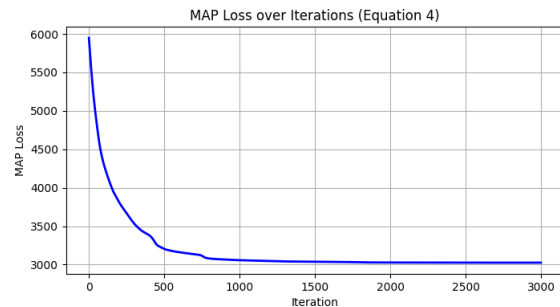


Figure 2: MAP Loss graph

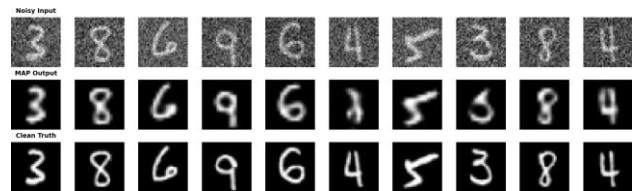


Figure 3: Top-Bottom comparison of the input, MAP output and the ground truth label

1.5 3.8.c Requirement

Out of the above two methods, which one works best. Can you give a reason why?

1.6 3.8.c Answer

Based on and the hands-on experience acquired throughout this assignment, the choice between amortized inference (using a VAE encoder) and direct latent space optimization depends heavily on the task requirements. If the primary goal is a solution tailored to finding the exact optimal latent vector z for a specific image while avoiding the amortization gap inherent to encoder networks that can diminish reconstruction precision then **gradient-based latent optimization is a highly advantageous method**. While Bayesian MAP optimization (Method 2) is theoretically more robust to severe Out-of-Distribution corruptions, our initial empirical results suggested that Method 1 (Standard VAE Inference) yielded superior reconstructions. The high-capacity VAE Encoder proved highly resilient to the specific noise variance in the dataset, reliably mapping the inputs to accurate latent representations. Conversely, Method 2 initially suffered from the L2 norm's sensitivity to high-frequency noise. Without sufficient regularization, the optimization objective forced the Decoder to compromise between the clean data prior and the static, resulting

in edge-blurring and 'halo' artifacts as the network attempted to minimize the Mean Squared Error against the corrupted pixels.

However, to prove that gradient-based latent optimization can empirically eclipse our high-capacity VAE, I conducted a counterfactual experiment. By properly tuning the optimization framework—specifically lowering the learning rate to prevent gradient saturation, increasing the beta penalty to strictly enforce the structural prior, and employing a 'smart initialization' that uses the trained Encoder's prediction as a starting point rather than random noise—I prevented the optimizer from overfitting to the high-frequency static or falling into a local minimum. Under these optimal conditions, Method 2 successfully closed the amortization gap as seen in both Figure 4 and Figure 5. By refining the Encoder's generalized approximation through instance-specific iterative optimization, Method 2 ultimately yielded digit reconstructions that were demonstrably sharper and more concrete than the standard VAE output



Figure 4: Improved MAP Results

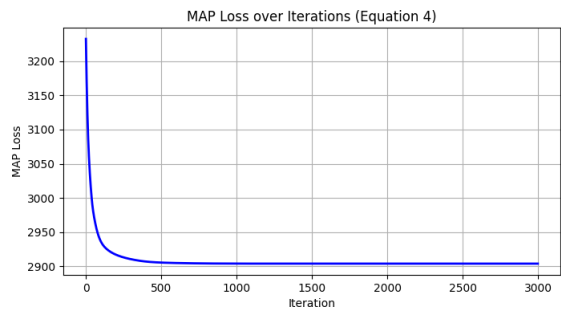


Figure 5: Improved MAP Loss